

# Cold Referee Report: Decidability of $\mathcal{ALCI}_{\text{RCC5}}$ Concept Satisfiability

Packet 1 of 3: Verdict and Executive Summary

Eighth review of the split-forest stack (first cold review of the round-14 effective stack)

July 9, 2026

**Reviewed materials.** (1) *Base*: “Split-Forest Automata” with appendices A–V (round 12). (2) *Delta 1*: round-13 report and repair (D-discipline, shadow amalgamation, width Section 8). (3) *Delta 2*: round-14 repaired proof (active virtual bags AV1–AV6). Effective stack: (1), as amended by the surviving parts of (2) — in particular its width-accounting Section 8 — as amended by (3); where (2) and (3) conflict, (3) governs.

**Reviewer independence note.** This review was conducted cold, with no carried-over project context. For transparency: the round-13 delta was authored by a model of the same family as this reviewer. The findings below are distributed across all three layers without regard to authorship; in particular, the surviving round-13 width section is itself faulted (Finding F3).

## 1 Verdict

**Question 1. Is the decidability theorem true?**

**Assessment: plausibly true; not settled by this stack, and not settled by this review.**

No counterexample to the theorem was found. The semantic core of the round-14 repair — active-virtual-bag amalgamation (Theorem 3.3), coverage (Lemma 4.1), the truth lemma (Theorem 4.3), and the set-level extraction (Theorem 5.1) — survived adversarial probing, including probes specifically targeting strong equality, witness reuse under forced uniqueness, and infinite deferral (Packet 3, Section C). The coarseness of RCC5 (machine-verified Facts 2.1–2.4, the patchwork property verified exhaustively in the small) removes the standard grid-encoding tools. However, the truth of the theorem currently rests on whether a genuinely *finite* decision layer exists for the AV abstraction, and that is exactly the part of the stack that is unproved (Finding F1). I regard the theorem as open.

**Question 2. Does the effective stack prove it?**

**Verdict: GAP — the stack does not prove decidability.** Two load-bearing obligations are unproved, and one of them is a structural open problem rather than deferred bookkeeping.

- **F1 (critical).** Theorem C — the finite two-way alternating parity automaton — has *no proof* in the effective stack for the round-14 abstraction language, and the monitor design sketched in round-14 Section 6 is unsound as described. A valid AV abstraction is no longer a labelled tree over a finite alphabet: it is a tree *plus* an unbounded enrichment  $N^*$  (a relation from every universal source to every port of every bag, and between every pair of

universal sources). This enrichment is not carried by the input tape; the sketch has it guessed by alternating run copies. Independent run copies cannot be forced to guess consistently, and I exhibit a concrete two-port configuration (machine-checked composition arithmetic:  $\text{Comp}(\text{PP}, \text{PP}) \cap \text{Comp}(\text{PP}, \text{DR}) = \{\text{PP}\} \cap \{\text{DR}\} = \emptyset$ ) in which every monitor described in Section 6 passes all of its local checks while *no coherent  $N^*$  exists at all*, i.e. acceptance without validity. All three natural repair routes (tape annotation, profile quotient, canonical guessing) are blocked by facts the stack itself establishes. Round-12’s own Theorem-C proof (Lemmas 5.2/5.3) is void because its validity clauses have been rewritten twice; no analogue is restated. Corollary 8.2 (decidability) therefore does not follow. Details: Packet 2, Finding F1.

- **F2 (major).** The width bound is never shown to absorb request-source liveness (D4). Round-13 Section 11(G2) explicitly named *two* width-pressure mechanisms — the D2 verticalization fallback and D4 liveness. Round-14 (Corollary 5.2) eliminates the first and is silent on the second, although the round-14 truth lemma makes D4 strictly load-bearing: under AV, non-universal sources have no active rows, so a deferred existential can only be discharged as a *live bag fact*, forcing the source to remain co-bagged along its support prefix. No lemma bounds prefix lengths or the number of simultaneously pending live sources. Details: Packet 2, Finding F2.
- **F3 (major).** The width bound’s provenance is internally contradictory and its surviving proof is not rigorous. Round-14 Section 1 declares itself a “replacement for round-13 Sections 3–8,” yet round-14 Theorem 5.1 relies on the bound  $K_A(C_0)$  that is defined and proved *only* in round-13 Section 8. As literally written, the effective stack either withdraws its only width-bound proof or the patch statement is wrong. Independently, round-13 Lemma 8.2’s recurrence does not close its own accounting: (g2)-generated ports are themselves “already generated” and may spawn further (g2) generation at the same rank; the one-sentence closure claim does not bound same-rank cascades. Details: Packet 2, Finding F3.

Minor findings F4–F7 (statement/use mismatch in Lemma 3.2, compactness wording in Theorem 3.3, a broken citation chain into round-12 Appendix H.2, and non-reproducibility of the WP scripts) are detailed in Packet 3; none is independently fatal, and for F4 and F5 I verified concrete repairs.

**Confidence.** High (approximately 0.9) that F1 is a genuine gap in the stack as written; this is not a judgment call — the text itself concedes in Remark 6.1 that the spawning discipline is unspecified, and I go further by showing the natural reading is unsound, not merely unwritten. Moderate (0.6–0.7) that closing F1 requires a substantially new idea (a bounded-interface lemma or an architecture change) rather than routine bookkeeping. High that F2 and F3 are unproved obligations as written; moderate that both are repairable with known techniques.

## 2 The effective stack, as reconstructed

Round 14 withdraws, from round 13: the horizontal-row restriction D1, the verticalization fallback D2, the pair-coherence clause D5 as a separate axiom, and the global DR default (clause (iii) of  $\Lambda$ ); D3 (type propagation) is retained but demoted to non-load-bearing. It replaces them by the active-virtual-bag discipline: every universal source is active at every bag ( $A_t = P_t \cup \text{Univ}_T$ , possibly infinite), with a coherent completion  $N_t^*$  subject to AV1–AV6. From round 12 it retains: Theorem A (split-forest normal form with thresholds and thinning), the bag/overlap infrastruc-

ture, strong-equality handling (Appendices G, R), witness/request handling for existentials, and the RCC5 patchwork theorem (Theorem 2.5, an explicitly external input). My audit of this discharge relation (Packet 3, S8) finds it complete *at the semantic level* with exactly two exceptions: the width section (F3) and the automaton (F1). One silent dependency break was found: round-12 Appendix H.2 sourced shadow rows from data “declared in the adjacent bag label,” a mechanism that no longer exists in the effective stack (F6, folded into the F1 narrative).

The decisive structural observation is this. Round 12 kept per-bag shadow data finite by recording rows in labels, but then could not have recorded rows for unboundedly many sources (the tension that produced rounds 13 and 14). Round 13 kept per-bag data bounded by *restricting* the language of rows (D1) and proving an existence lemma (Fact 2.1) for everything unrecorded — and broke on width (G2). Round 14 makes the data total and the existence lemma trivial — and thereby moves the entire difficulty into the automaton, where it now sits unproved. The gap has migrated across three rounds; it has not closed.

### 3 Ground truth verified by computation

All of the following were verified by a fresh script written for this review (source and outputs in Packet 3, Section D); none is taken from the manuscripts’ WP transcripts.

| Item                                                                                                                                 | Result                             |
|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| RCC5 composition table derived from set semantics (63 regions over a 6-element universe, all $6^3$ triples) vs. base Section 2 table | exact match, all 25 cells          |
| Fact 2.1 (horizontal absorption), Fact 2.2 (vertical transitivity), Fact 2.3 (no horizontal dead ends), Fact 2.4 (EQ never forced)   | all hold                           |
| WP20 triangle: $PO \in \text{Comp}(PP, DR)$ ?                                                                                        | no; $\text{Comp}(PP, DR) = \{DR\}$ |
| Closed complete atomic networks                                                                                                      | 41 ( $n=3$ ), 916 ( $n=4$ )        |
| Patchwork, 3+3 over a 2-overlap (all 427 agreeing pairs)                                                                             | 0 failures                         |
| Patchwork, 4+4 over a 3-overlap (all 21,094 agreeing pairs)                                                                          | 0 failures                         |
| DR-isolation extension lemma (fresh point DR to all), $n \leq 4$                                                                     | 0 failures                         |
| F1 witness arithmetic: $\text{Comp}(PP, PP) \cap \text{Comp}(PP, DR)$                                                                | $\emptyset$                        |
| Source-triangle necessity: $DR \in \text{Comp}(PP, PP)$ ?                                                                            | no                                 |

The composition table is taken as ground truth for everything below, per instructions. The patchwork property (Theorem 2.5) is required for *all* finite sizes; my exhaustive checks cover the two smallest nontrivial amalgamation shapes and are supporting evidence only — I follow the base manuscript in treating the general statement as an external input, and flag (Packet 3, S9) that the stack’s reliance on it should continue to be stated prominently.

## 4 Summary table of findings

| ID | Severity | One-line statement                                                                                                              | Where    |
|----|----------|---------------------------------------------------------------------------------------------------------------------------------|----------|
| F1 | Critical | Theorem C unproved for the AV language; Section-6 monitor sketch unsound as described (copy-consistency of guessed active data) | Packet 2 |
| F2 | Major    | Request-source liveness (D4) never integrated into the width bound; round-13 G2 only half-discharged                            | Packet 2 |
| F3 | Major    | Width bound $K_A(C_0)$ rests on a nominally withdrawn section, whose Lemma 8.2 recurrence does not close                        | Packet 2 |
| F4 | Minor    | Lemma 3.2 statement/use mismatch; repair verified (DR-isolation)                                                                | Packet 3 |
| F5 | Minor    | Theorem 3.3 compactness wording ( $C_F$ /completion quantification); repair verified                                            | Packet 3 |
| F6 | Minor    | Broken citation chain: round-12 App. H.2’s label-declared rows no longer exist                                                  | Packet 3 |
| F7 | Minor    | WP scripts not included; machine-check outputs unproducibile as submitted                                                       | Packet 3 |

Steps checked and found sound (S1–S10), the log of adversarial probes attempted and absorbed (p1–p7), and recommendations for a round 15 (R1–R6) are in Packet 3.

## 5 Method statement

Verified by actual computation: everything in the table of Section 3. Verified by proof reconstruction (I re-derived the argument and checked each step against the ground-truth table): Lemmas 3.1, 3.2, Theorem 3.3, Lemma 4.1, Theorem 4.3, Theorem 5.1 at the set level; the AV3 global-rigidity chain; the bidirectional coverage argument behind AV4; the strong-equality path end to end; the discharge audit of Section 2 above. Read-through with targeted spot checks: Theorem A and its appendices (previously reviewed material, reported stable), the parity layer, the alphabet cardinality claims. Wherever the text asserts a step as “clear” that I could not reconstruct, it appears above as a finding, per the review instructions; the three such places are exactly F1, F2, and F3.

*End of Packet 1. Packet 2 contains the major findings in full detail with witnesses; Packet 3 contains verified steps, minor findings, the probe log, the computational appendix, and recommendations.*